

PageRank Analysis Using Iterative and Eigenvalue Methods on Directed Web Graphs

LILY NGUYEN¹ AND ANDRE SAE¹

¹*Department of Physics, The University of Texas at Austin
Austin, TX 78712, USA*

ABSTRACT

In this project, we explore numerical methods for computing the PageRank vector, which ranks web pages based on their relative importance. The PageRank matrix is a stochastic transition matrix that encodes the probability of moving from one page to another, and its dominant eigenvector represents the steady-state distribution of page visits (PageRank score). We first implement an iterative method that repeatedly applies the hyperlink matrix $\mathbf{H}$ to a uniform probability vector and observe whether it converges to a steady-state solution. We then apply a direct method by solving for the eigenvector corresponding to the eigenvalue $\lambda = 1$, known as the dominant eigenvector. Both methods assume the matrix is ergodic to ensure convergence. When this condition fails, such as in the presence of dangling nodes or disconnected pages, we introduce the "random surfer" model, which adds a damping factor that enables jumps to any node to ensure convergence and reflect more realistic web navigation.

Keywords: stochastic matrix — hyperlink — directed graph — PageRank — eigenvalue — eigenvector — internet — search engine

1. INTRODUCTION

The PageRank algorithm, first introduced by Larry Page and Sergey Brin, was one of the first algorithms to rank webpages based on their link structure rather than solely content [Brin & Page \(1998\)](#). The algorithm models the web as a directed graph and simulates an "agent" who follows hyperlinks with some probability. Essentially, it tracks how often this "agent" would land on a particular page over an extended period, so pages that are visited more frequently are considered more important and thus receive a higher PageRank score [Page et al. \(1999\)](#).

This setup defines a Markov process where each page's importance is proportional to the importance of pages linking to it. The final PageRank vector corresponds to the steady-state distribution of this process and can be found either by iteration or by solving for the dominant eigenvector of the hyperlink matrix $\mathbf{H}$ [Langville & Meyer \(2009\)](#).

In this project, we first implemented PageRank using the difference equation $\mathbf{x}_{k+1} = \mathbf{H}\mathbf{x}_k$, which resulted in the scores converging once the solution reached the maximum tolerance allowed. We then confirmed our result by solving for the eigenvector associated with

the largest eigenvalue of the matrix $\mathbf{H}$. Our goal is to understand how the algorithm behaves under different network structures and how quickly it converges [Langville & Meyer \(2004\)](#).

2. DATA AND OBSERVATIONS

For an $n \times n$ hyperlink matrix $\mathbf{H}$ to be stochastic, each column must be a probability vector with non-negative entries that sum to 1. The matrix must also be ergodic, it is both irreducible and aperiodic. This ensures that probability is conserved at each step of the Markov process and allows the system to converge to a unique steady-state solution. This condition is only satisfied if every page has at least one outgoing link. Otherwise, the corresponding column in $\mathbf{H}$ would contain all zeroes and violate the stochastic property [Langville & Meyer \(2009\)](#).

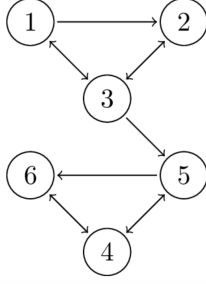


Figure 1. Example web graph.

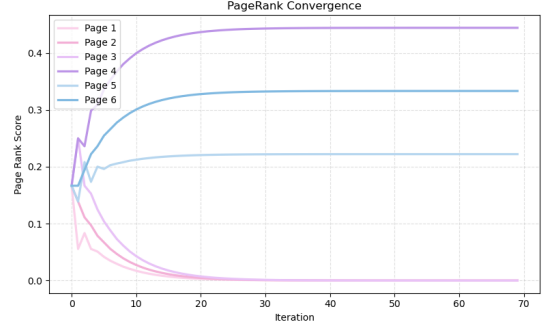


Figure 2. PageRank convergence for graph 1.

To start, we defined a 6-node directed graph where each page links to one or more pages, as shown in Figure 1. From this, we constructed the hyperlink matrix $\mathbf{H}$, where each non-zero entry $\mathbf{H}_{ij}$ is equal to $\frac{1}{|P_{ij}|}$, the reciprocal of the number of outbound links from page P_j . This matrix represents the probability that an agent on page j clicks through to page i :

$$\mathbf{H} = \begin{pmatrix} 0 & 0 & 1/3 & 0 & 0 & 0 \\ 1/2 & 0 & 1/3 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/2 & 1 \\ 0 & 0 & 1/3 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 \end{pmatrix} \quad (1)$$

We initialized the PageRank vector $\mathbf{x}_0$ with equal values and assumed each page is equally likely to be visited at first:

$$\mathbf{x}_0 = \begin{pmatrix} 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \end{pmatrix} \quad (2)$$

We applied the iterative update rule $\mathbf{x}_{n+1} = \mathbf{H}\mathbf{x}_n$ until the rank vector converged below a given tolerance to simulate an agent moving through the web graph via the links. We visualized the convergence of the PageRank vector by plotting each page's rank score for each iteration, which allowed us to track how the scores changed until they reached a steady state. Our results are shown in Figure 2.

The convergence of the iterative method is a direct consequence of the underlying theory of stochastic matrices. According to Theorem 4.9, if $\mathbf{H}$ is an $n \times n$ stochastic matrix and $\mathbf{x}_0$ is some initial vector, then the system defined by the difference equation $\mathbf{x}_{n+1} = \mathbf{H}\mathbf{x}_n$ converges to a unique steady-state vector [Arasu et al. \(2002\)](#). This limit corresponds to the eigenvector $\mathbf{v}_1$ associated with eigenvalue $\lambda = 1$, since all the other eigenvalue components vanish as $k \rightarrow \infty$:

$$\mathbf{x}_{\text{equilibrium}} = \lim_{k \rightarrow \infty} \mathbf{H}^k \mathbf{x}_0 = c_1 \mathbf{v}_1 \quad (3)$$

Theorem 4.9 greatly simplifies the PageRank iterative process. Instead of repeatedly applying $\mathbf{x}_{n+1} = \mathbf{H}\mathbf{x}_n$, we can directly compute the dominant eigenvector of $\mathbf{H}$ corresponding to the eigenvalue $\lambda = 1$. This vector represents the steady-state solution and results in the same PageRank distribution as the limit of the iterative method.

Both the iterative difference equation and the eigenvector method yield the same steady-state PageRank vector:

$$\mathbf{x}_{\text{converged}} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 4/9 \\ 2/9 \\ 1/3 \end{pmatrix} \quad (4)$$

We applied the PageRank algorithm to a second web graph consisting of eight interconnected pages, as shown in Figure 3.

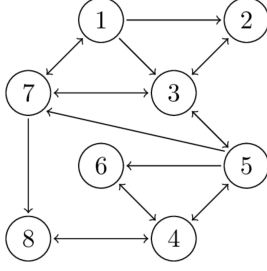


Figure 3. Example web graph.

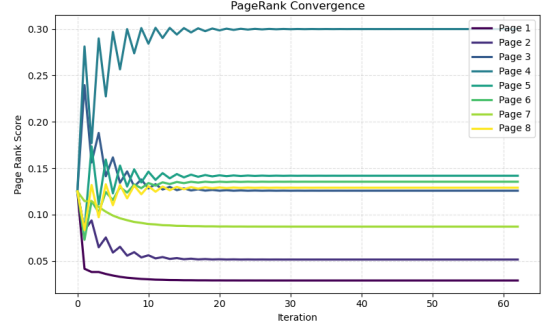


Figure 4. PageRank Convergence for graph 2.

101 The corresponding hyperlink matrix $\mathbf{H}$ and initial uni-
102 form state vector $\mathbf{H}$ are defined below.

$$\mathbf{H} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1/3 & 0 \\ 1/3 & 0 & 1/3 & 0 & 0 & 0 & 0 & 0 \\ 1/3 & 1 & 0 & 0 & 1/4 & 0 & 1/3 & 0 \\ 0 & 0 & 0 & 0 & 1/4 & 1 & 0 & 1 \\ 0 & 0 & 1/3 & 1/3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 1/4 & 0 & 0 & 0 \\ 1/3 & 0 & 1/3 & 0 & 1/4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 0 & 0 & 1/3 & 0 \end{pmatrix} \quad (5)$$

$$\mathbf{x}_0 = \begin{pmatrix} 1/8 \\ 1/8 \\ 1/8 \\ 1/8 \\ 1/8 \\ 1/8 \\ 1/8 \\ 1/8 \end{pmatrix} \quad (6)$$

105 Using both the iterative update method and the dom-
106 inant eigenvector approach, we computed the steady-
107 state PageRank distribution. The convergence behavior
108 of the iterative method is shown in Figure 4

109 Using both the iterative difference equation and the
110 dominant eigenvector method, we arrive at the same
111 steady-state PageRank vector:

$$\mathbf{x}_{\text{converged}} = \begin{pmatrix} 0.0290 \\ 0.0516 \\ 0.1258 \\ 0.3000 \\ 0.1419 \\ 0.1355 \\ 0.0871 \\ 0.1290 \end{pmatrix} \quad (7)$$

3. RESULTS

The distribution for graph in Figure 1 reflects the long-term behavior of an agent navigating the web graph. Based on the values, we rank the pages in order of importance as follows:

Page 4 > Page 6 > Page 5 > Page 1 = Page 2 = Page 3

116 After a long enough time span, $\frac{4}{9}$ of users will end up in
117 Page 4, $\frac{2}{9}$ of users will end up in page 5, and $\frac{1}{3}$ of users
118 will end up in Page 1. No one will end up on Page 1, 2,
119 and 3.

From the results of the graph in Figure 3, we rank the pages in order of importance:

Page 4 > Page 5 > Page 6 > Page 8 > Page 3
> Page 7 > Page 2 > Page 1

120 Page 4 is the most dominant, whereas Page 1 has the
121 lowest influence in the network. In other words most
122 people will end up on page 4 and the least amount of
123 people will end up on page 1 given a long enough times-
124 pan.

4. SUMMARY AND CONCLUSION

To summarize, we implemented two numerical approaches to compute the PageRank vector. The first was an iterative method that started from a uniform initial distribution $\mathbf{x}_0$ and repeatedly applied the update equation $\mathbf{x}_{n+1} = \mathbf{H}\mathbf{x}_n$ until the difference between successive iterations fell below a convergence threshold. The second approach used Theorem 4.9, which states that the steady-state distribution is the dominant eigenvector of the stochastic matrix $\mathbf{H}$. Both methods gave identical results, which confirms the accuracy of our implementation to describe long term internet behavior. We performed all computations in Python using NumPy and visualized convergence behavior with matplotlib.

One limitation of our PageRank model implementation is that we assumed each transition to be independent of each other, meaning the current and previous pages visited do not influence the next page. While this may have been reasonable when the internet was more decentralized, it does not reflect the behavior driven by modern recommendation algorithms, which actively shape user navigation to maximize engagement and retention.

Another limitation we encountered is that the PageRank Matrix $\mathbf{H}$ must be ergodic for the iterative method to

converge. In cases where this condition fails, we can introduce a perturbation method to enforce ergodicity. This can be achieved by modifying the original matrix with a damping factor α , where n is a ones matrix and n is the total number of pages:

$$\hat{H} = (1 - \alpha)H + \frac{\alpha}{n}J \quad (8)$$

This perturbation represents the probability that a user may randomly jump to any page and ensures the matrix remains stochastic to prevent the system from becoming trapped in cycles Boldi et al. (2005). Thus, the random surfer model not only guarantees convergence, but also aligns more closely with realistic user behavior in the modern web.

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